

IAL Biology Unit 1 (WBI11): Rapid Revision

Molecules, Diet, Transport and Health · last-minute summary · 1 h 30 min · 80 marks · 8 questions, no multiple-choice section · marks ramp from ~6 (Q1) to ~14 (Q8) · Q5-Q8 hold about 62% of the paper

HOW THE MARKS WORK

- **“Explain” is the dominant command** — about 9.6 marks a paper. It needs a **causal chain**, not a list: link with *because, so, therefore, which means*.
- **Calculations in all 22 papers**, averaging 3.7 marks. Always show working and give units.
- **Guaranteed every session:** maths and data skills, carbohydrates, gametes and fertilisation, cell ultra-structure.
- **Near-certain:** circulation (18/22), diet and cholesterol (18), proteins (17), enzymes (16), atherosclerosis (16).
- **Barely tested:** evolution; formal statistical tests.
- Quote **figures with units** whenever you describe data.

CARBOHYDRATES

Molecule	Structure and role
Monosaccharide	Glucose $C_6H_{12}O_6$; α (OH below C1) and β (OH above)
Disaccharide	Condensation, glycosidic bond , water released. Maltose = 2 glucose; sucrose = glucose + fructose; lactose = glucose + galactose
Starch	Amylose: α -1,4, unbranched, helical, compact . Amylopectin: 1,4 + 1,6 branches → many ends → rapid hydrolysis
Glycogen	Like amylopectin but more branched → faster glucose release; animal store
Cellulose	β -glucose, alternate molecules inverted , straight chains, H-bonded into microfibrils → tensile strength

Why storage polysaccharides are good: **insoluble** (no osmotic effect), compact, easily hydrolysed, large so they do not leave the cell.

LIPIDS AND PROTEINS

- **Triglyceride** = 1 glycerol + **3 fatty acids**, **ester bonds**, condensation, **3 waters released**.
- **Saturated:** no C=C, straight chains, **pack closely**, solid at room temperature. **Unsaturated:** one or more C=C → **kinks** → **pack less closely** → liquid. Mono = 1 double bond, poly = more than 1.
- More energy per gram than carbohydrate — **more C-H bonds, less oxygen** (more reduced).
- **Phospholipid:** hydrophilic phosphate head + 2 hydrophobic tails → forms a bilayer.

Amino acid = central C + **-NH₂** + **-COOH** + H + **R group**. Joined by condensation → **peptide bond**.

Level	What it is, and the bonds
Primary	Sequence of amino acids; peptide bonds
Secondary	α -helix / β -pleated sheet; H bonds between C=O and N-H of the backbone
Tertiary	3D folding of one chain; H bonds, ionic, disulfide bridges , hydrophobic — all between R groups
Quaternary	More than one polypeptide chain

Globular: compact, spherical, hydrophilic R groups outward, **soluble** — enzymes, haemoglobin, antibodies.

Fibrous: long parallel chains, **insoluble**, structural — collagen, keratin.

Collagen: 3 chains in a **triple helix**, H bonds, **glycine every third amino acid** (small enough for the centre), **cross-links** and **staggered ends** → tensile strength.

ENZYMES

- Globular proteins, biological catalysts, **lower the activation energy**.
- **Active site complementary** to substrate → **enzyme-substrate complex**.
- **Lock and key** = exact fit already. **Induced fit** = active site **changes shape slightly** on binding, straining substrate bonds.

Factor	Explanation that scores
Temperature ↑	More kinetic energy → more frequent successful collisions → more E-S complexes. Above the optimum: H and ionic bonds holding the tertiary structure break → active site changes shape → no longer complementary → denatured
pH	R groups change charge → ionic and H bonds disrupted → active site shape altered
[Substrate] ↑	Rate rises until all active sites are occupied (saturated) → plateau at V_{max} ; enzyme concentration now limiting
Competitive inhibitor	Similar shape → binds the active site → effect reduced by raising [substrate]
Non-competitive	Binds elsewhere → changes active site shape → not overcome by raising [substrate]

Never write “the enzyme is killed / destroyed / melts”. Denaturation breaks the bonds maintaining **tertiary** structure — **not** peptide bonds. Primary structure is unchanged.

HEART AND CARDIAC CYCLE

1. **Atrial systole:** atria contract, pressure rises above ventricular, AV valves open, ventricles fill.
2. **Ventricular systole:** ventricles contract, pressure exceeds atrial → **AV valves close**; exceeds arterial → **semilunar valves open**, blood ejected.
3. **Diastole:** all relax, ventricular pressure falls below arterial → semilunar valves close; below atrial → AV valves open.

Valves open and close because of **pressure differences** — say which chamber is higher. **Left ventricle wall is thicker** because it pumps blood **to the whole body** against higher resistance; the right pumps only to the lungs, at lower pressure, so capillaries are not damaged.

Cardiac output = heart rate × stroke volume. Heart rate from a trace = $60 \div \text{time for one cycle}$.

Vessel	Structure related to function
Artery	Thick elastic and muscular wall, narrow lumen — withstands and smooths high pressure (recoil)
Capillary	One cell thick, narrow lumen , pores — short diffusion distance, large surface area
Vein	Thin wall, wide lumen, valves — low pressure, prevents backflow

CARDIOVASCULAR DISEASE

ATHEROSCLEROSIS: THE CHAIN, IN ORDER

1. Endothelium damaged (high blood pressure, toxins from smoking). 2. **Inflammatory response**; white blood cells accumulate. 3. Cholesterol and lipids deposit → **atheroma**. 4. Fibrous tissue and calcium salts → hard **plaque**. 5. Lumen **narrows**, artery **loses elasticity** → blood pressure rises further → positive feedback. 6. Reduced blood flow → less oxygen to tissue; plaque may rupture → **thrombosis**.

Blood clotting: damage → platelets and damaged tissue release **thromboplastin** → (with Ca^{2+} and vitamin K) **prothrombin** → **thrombin** → thrombin catalyses **soluble fibrinogen** → **insoluble fibrin** → mesh traps platelets and red cells → clot.

Risk factor	Mechanism
High blood pressure	Damages the endothelium
Smoking	CO reduces oxygen transport; nicotine raises heart rate and causes vasoconstriction; damages endothelium
High blood cholesterol	LDLs deposit in artery walls
Diet high in saturated fat	Raises LDL
Obesity, inactivity, diabetes, age, sex, genetics	Non-modifiable ones cannot be changed — say so if asked

LDL carries cholesterol **to** tissues, binds receptors, deposits in artery walls — raised by saturated fat. **HDL** carries cholesterol **to the liver** for breakdown — raised

by unsaturated fat.

EVALUATING RISK-FACTOR DATA

Say **correlation does not prove causation**; name an uncontrolled variable; comment on **sample size**, length of study, whether it is representative, and whether the difference is **significant**. Quote figures from the data with units.

MEMBRANES AND TRANSPORT

Fluid mosaic: phospholipid bilayer with hydrophilic heads outward and hydrophobic tails inward; proteins (channel, carrier, intrinsic, extrinsic); cholesterol regulates **fluidity**; glycoproteins and glycolipids for recognition. “Fluid” = molecules move laterally; “mosaic” = scattered proteins.

Process	Key points
Diffusion	Down a gradient, passive, small non-polar molecules
Facilitated diffusion	Down a gradient, passive, channel or carrier proteins , for polar or charged particles
Osmosis	Movement of water from higher to lower water potential through a partially permeable membrane
Active transport	Against the gradient, carrier protein, ATP , protein changes shape
Exo- / endocytosis	Bulk transport by vesicles; requires ATP

Fick's law: $\text{rate} \propto (\text{surface area} \times \text{concentration difference}) \div \text{thickness}$. Name all three terms when explaining a gas-exchange surface.

DNA AND PROTEIN SYNTHESIS

Nucleotide = pentose sugar + phosphate + base. DNA: deoxyribose, double helix, antiparallel, **A-T (2 H bonds)**, **C-G (3)**, phosphodiester bonds. RNA: ribose, single strand, uracil.

Semi-conservative replication: DNA helicase breaks the **hydrogen bonds** → strands separate → each acts as a **template** → free nucleotides pair by **complementary base pairing** → **DNA polymerase** joins them with phosphodiester bonds → each new molecule has **one original and one new strand**.

Transcription: in the nucleus; one strand is the template; **RNA polymerase**; mRNA leaves through a nuclear pore. **Translation:** at the ribosome; **tRNA anticodon** pairs with the **mRNA codon**; amino acids joined by peptide bonds; the **genetic code is degenerate** (more than one codon per amino acid), non-overlapping and universal.

MUTATION AND CYSTIC FIBROSIS

A change in the **base sequence** → a different amino acid sequence → different **bonds between R groups** → different **tertiary structure** → protein does not function.

CF: a **deletion of three bases** → one amino acid missing from **CFTR**, a chloride channel. Chloride ions are not secreted → water does not follow by osmosis → **mucus is thick and sticky**. Lungs: blocks airways, traps bacteria → infection; cilia cannot clear it. Pancreas: blocks the duct →

enzymes do not reach the gut → poor digestion. Reproductive: blocks tubes → reduced fertility.

Genetic screening terms: **amniocentesis** (15–17 weeks, amniotic fluid, ~1% **miscarriage risk**, later result); **CVS** (8–12 weeks, placenta, slightly higher risk, earlier result); **PGD** with IVF. Ethical answers must give **both sides** and avoid asserting a single right answer.

GAMETES AND GENETICS

Sperm	Egg
Acrosome with hydrolytic enzymes to digest the zona pellucida	Zona pellucida ; cortical granules
Many mitochondria for ATP for the tail	Food reserves in the cytoplasm
Haploid nucleus; flagellum	Haploid nucleus; large

Fertilisation: sperm binds the zona pellucida → **acrosome reaction** releases enzymes digesting a path → sperm membrane fuses with the egg membrane → nucleus enters → **cortical reaction:** cortical granules release contents that thicken the zona pellucida, **preventing polyspermy** → nuclei fuse → diploid zygote.

Genetics: define the allele symbols, draw the full **Punnett square** with parental genotypes and gametes, and **state the ratio**. Autosomal recessive (CF, albinism): carriers are heterozygous; two carriers → 1 in 4 affected. Sex-linked: write alleles on the X, as X^HY .

MATHS AND DATA SKILLS

Present in all 22 papers. % change = $(\text{new} - \text{original}) \div \text{original} \times 100$ · rate = $\text{change} \div \text{time}$ **with units** · magnification = $\text{image} \div \text{actual}$ (convert to the same unit first) · mean, median, mode · **standard deviation** shows spread; **overlapping error bars** → **not significant**, and you must **name the groups**. Give answers to a sensible number of significant figures and always include the unit.

THE REJECTIONS BANK

Do not write	Write instead
Enzyme is “killed”, “destroyed”, “melts”	Denatured: bonds holding the tertiary structure break, active site changes shape
“Peptide bonds break” on denaturation	Hydrogen and ionic bonds break; primary structure is unchanged
“The enzyme fits the substrate”	The active site is complementary to the substrate
“Capillaries dilate”	Vasodilation of arterioles / blood flow increases
“Cholesterol blocks the artery”	Describe the full atheroma → plaque → narrowed lumen chain
“Correlation shows X causes Y”	Correlation does not prove causation ; name an uncontrolled variable
“Water moves to where there is more solute”	Water moves from higher to lower water potential
“DNA splits in half”	Hydrogen bonds break; semi-conservative replication
“The gene makes the protein”	The base sequence determines the amino acid sequence , hence the tertiary structure
Figures with no units	Always quote figures with units

WALKING INTO THE EXAM

Circle the command word and the mark allocation. “Explain” needs a **linked chain**; six unlinked facts score four, not six. Quote data with units. On the later, heavier questions, give one distinct point per mark. Never leave a calculation blank — method marks are awarded on working.